19 Hypothesis Testing 2

Ex 1 Company claims their cars can accelerate from 0 to 60 mph in 6 seconds. A random sample of 33 cars has a mean acceleration of 7.2 seconds. At $\alpha = 0.05$, can you reject the claim? (Assume pop. standard deviation of 2.5 seconds).

Hypothesis Testing for mean μ (σ unknown)

State $H_0: \mu = \mu_0$, $\mu \geq \mu_0$, $\mu \leq \mu_0$

$H_a: \mu \neq \mu_0$, $\mu < \mu_0$, $\mu > \mu_0$

Check 10% Condition : $N \geq 10n$

Normal / Large Sample : Pop. is normally distributed or $n \geq 30$

Compute P-value $P = pt\left(\frac{\bar{x} - \mu_0}{s_x / \sqrt{n}}, df = n-1\right)$ ... left-Tail

$P = pt\left(\frac{\bar{x} - \mu_0}{s_x / \sqrt{n}}, df = n-1, \text{lower.Tail} = \text{FALSE}\right)$ ... right-Tail

$P = 2 \cdot pt\left(\frac{\bar{x} - \mu_0}{s_x / \sqrt{n}}, df = n-1\right)$ ... Two-Tailed

Conclude : If $P \geq \alpha$, we fail to reject H_0. If $P < \alpha$, we reject H_0.

Ex 2 Used car dealer claims the mean price of used cars sold in the past 12 months is at least \$21,000. You find that a random sample of 14 used cars sold in the past 12 months have a mean price of \$19,200 and sd of \$3000. Is there enough evidence to reject the dealer's claim at $\alpha = 0.05$? Assume the pop. is normally distributed.

Hypothesis Testing for proportion p

State $H_0: p = p_0,\ p \geq p_0,\ p \leq p_0$

$H_a: p \neq p_0,\ p < p_0,\ p > p_0.$

Check 10% Condition: $N \geq 10n$

Large Counts: np_0 and $n(1-p_0)$ are ≥ 10.

Compute P-value $P = \text{pnorm}\left(\hat{p},\ \text{mean} = p_0,\ \text{sd} = \sqrt{\frac{p_0(1-p_0)}{n}}\right)$ ... left-Tail

$P = \text{pnorm}\left(\hat{p},\ \text{mean} = p_0,\ \text{sd} = \sqrt{\frac{p_0(1-p_0)}{n}},\ \text{lower.Tail} = \text{FALSE}\right)$ ... right-Tail

$P = 2 \cdot \text{pnorm}\left(\hat{p},\ \text{mean} = p_0,\ \text{sd} = \sqrt{\frac{p_0(1-p_0)}{n}}\right)$ ... Two-Tail

Conclude: If $P < \alpha$, we reject H_0. If $P \geq \alpha$, we fail to reject H_0.

Ex News claim $\geq 29\%$ of US employees have changed jobs in the past 3 years. In a random sample of 180 US employees, 43 have changed jobs in the past 3 years. At the $\alpha = 0.10$ level, is there enough evidence to counter the news' claim?